

PRA PSPM
DC014 COMPUTER SCIENCE
2024/2025
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INSTRUCTION:
ANSWER ALL THE QUESTIONS.

- 1** E-mail, Yahoo Messenger, Video Conferencing and Multimedia Message Service (MMS) are examples of multimedia applications.

- (a) Describe **one (1)** most attractive multimedia element, that could be include in the applications.

[2 marks]

- (b) Explain the purpose of most attractive multimedia element mentioned in 1(a).

[2 marks]

- (c) Compare **specifically** the difference between three-dimensional (3D) and four-dimensional (4D) in multimedia application representation in terms of dimension.

[2 marks]

	Three-dimensional (3D)	Four-dimensional (4D)
Dimension		

- 2** The elegance and simplicity of the three-phase (3Ps) model have contributed to its widespread adoption as a conceptual framework for the production process.

- (a) Briefly describe the second phase of the multimedia design.

[2 marks]

- (b) Describe **one (1)** activity in the first and third phase of the multimedia design.

[6 marks]

First Phase:	Third Phase:
Activity:	Activity:

- (c) An authoring tool is a program that enables you to create a digital course and publish it in a select format.
- (i) Differentiate among two types of multimedia authoring tools shown below.

[2 marks]

Criteria	Icon-Based	Card/Page-Based
Technique		

(ii)

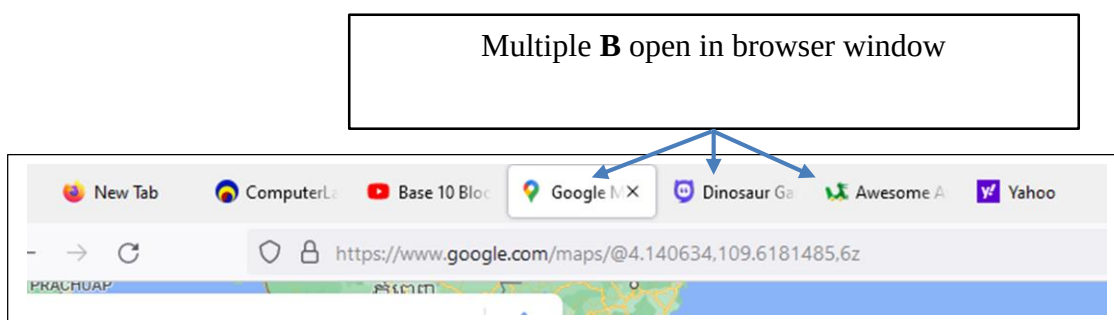


FIGURE 1

B is one of the types of multimedia navigation. Explain **B** in FIGURE 1.

[2 marks]

- (d) Give **two (2)** differences between offline and online multimedia delivery.

[4 marks]

Offline	Online

- 3 (a) Differentiate between **File Processing System** and the **Database Approach**.

[2 marks]

File Processing System	Database Approach

- (b) Describe **two (2)** downsides of using the database approach.

[2 marks]

- (c) Identify **two (2)** components of a database environment.

[2 marks]

- 4 A database management system (**DBMS**) is *software that allows users to create a computerized database; add, modify, and delete data in the database.*

column

Y

CustomerId	FirstName	LastName	DateCreated	Cli
1	Homer	Simpson	13/06/2014 3:33:37 PM	
2	Peter	Griffin	13/06/2014 9:09:56 PM	
3	Stewie	Griffin	13/06/2014 9:16:07 PM	
4	Brian	Griffin	13/06/2014 9:16:36 PM	
5	Cosmo	Kramer	13/06/2014 9:16:41 PM	

row

X

FIGURE 2

- (a) Describe the representations of a relational database in **FIGURE 2**.

[2 marks]

X	
Y	

- (b) Data integrity is a condition in which the data in the database complies with all entity and referential integrity constraints. Explain **one (1)** concept of data integrity.

[2 marks]

- 5 With a network, resources can be shared between different computers and this reduces the overall operating cost.

- (a) (i) State **two (2)** types of networking media.

[2 marks]

A	B

- (ii) Describe types of networking media mentioned in 5 (a).

[4 marks]

A	
B	

- (iii) Give **two (2)** examples of each of the networking media mentioned in 5 (a).

[4 marks]

Type	A	B
Example 1		
Example 2		

- (b) A computer network is a collection of computers and devices connected together via communication devices and transmission media.

- (i) Describe the function of switch in the network.

[2 marks]

- (ii) Explain why router is used in the network.

[2 marks]

- (iii) Besides switch and router, wireless access point and network interface card also network device that can used to build a network. Describe the function of each.

[4 marks]

Network device	Function
Wireless Access Point	
Network Interface Card	

6 (a)

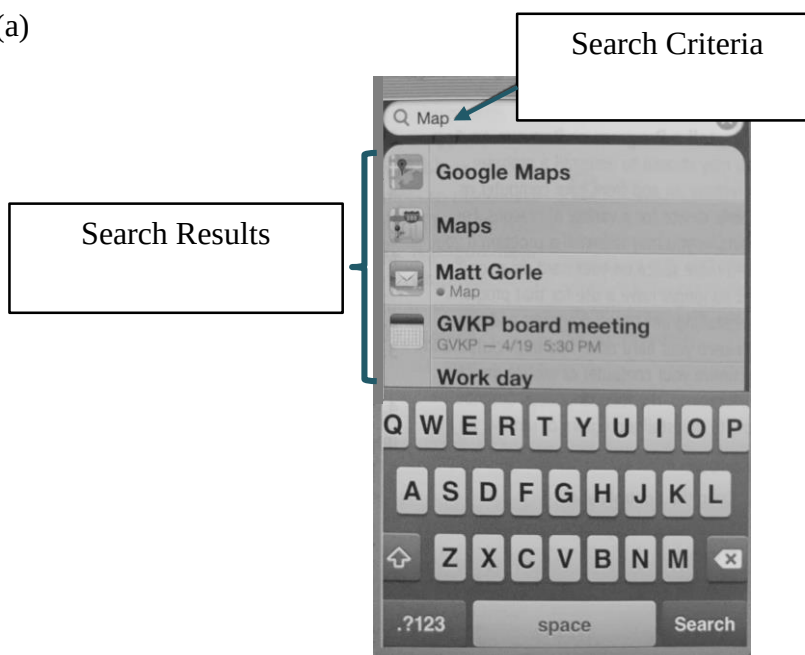


FIGURE 3

This search displays all files on the mobile device that match the search criteria, Map (**FIGURE 3**). Notice the search results show Map apps, email with a map, and a calendar event.

(i) Describe **two (2)** types of search tools.

[4 marks]

Search tool 1:	Search tool 2:

(ii) Give **one (1)** example for both search tools.

[2 marks]

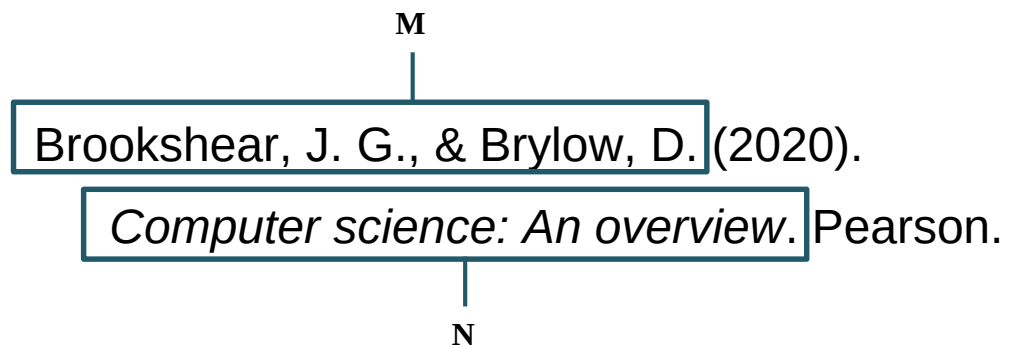
- (iii) Search strategies will help you to narrow down the list of hits to more manageable numbers so that the searching process can be more efficient and effective. List **two (2)** search strategies that you can use for better relevant web results.

[2 marks]

- (iv) Describe **one (1)** of search strategies that you mentioned in 6 (a) (iii) briefly.

[2 marks]

- (v) American Psychological Association (APA) is the examples of citing sources.



Identify the order of labels according to the APA system.

[2 marks]

M	
N	

- (b) Data privacy is the relationship between the collection and dissemination of data, technology, the public expectation of privacy, and the legal and political issues surrounding them.

Describe privacy in

- (i) Web browsing

[2 marks]

- (ii) Email privacy

[2 marks]

- (c) The Internet of Things (IoT) refers to a world where every day physical objects are connected to, and uniquely identifiable on, so they can communicate with other devices.

Explain **one (1)** personal application of IoT.

[2 marks]

END OF QUESTION PAPER